Machine Learning A (2026–2027)
Home Assignment 3

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1 Optimal Classification (16 points)

Question 1

I find the optimal deterministic classification rule by applying Bayes-optimal rule under 0-1 loss. This means choosing the value of $\hat{y}$ which maximizes the probability of $\hat{y} = y$ given the input variable x:

$$\hat{y}(x) = \arg\max_{y \in \{0,1\}} P(Y = y \mid X = x).$$

Applying this yields:

$$\hat{y} = 0 \quad \text{for all } x \in \{0, 1\}.$$

Assuming that $P(X = 0) = P(X = 1) = 0.5$, i get the expected 0-1 loss:

$$\begin{aligned}
\mathcal{R}_p(\hat{y}(X)) &= \mathbb{E}_X\left[P(Y \neq \hat{y}(X)) \mid X)\right], \\
&= P(X = 0) P(Y \neq \hat{y}(0))) + P(X = 1) P(Y \neq \hat{y}(1))), \\
&= 0.5 \cdot P(Y = 1 \mid X = 0) + 0.5 \cdot P(Y = 1 \mid X = 1), \\
&= 0.5 \cdot 0 + 0.5 \cdot 0.2, \\
&= 0.1.
\end{aligned}$$

Question 2

With the new probabilistic classifier:

$$\hat{y} = \begin{cases} 0 & \text{for } x = 0 \\ P(\hat{y} = 0) = 0.8 \text{ and } P(\hat{y} = 1) = 0.2 & \text{for } x = 1 \end{cases},$$

the expected 0-1 loss is:

$$\begin{aligned}
\mathcal{R}_p(\hat{y}(X)) &= P(X = 0) P(\hat{y} \neq Y \mid X = 0) + P(X = 1) P(\hat{y} \neq Y \mid X = 1), \\
&= 0.5 P(Y = 1 \mid X = 0) + \\
&\quad 0.5 \left(P(\hat{y} = 0 \mid X = 1) P(Y = 1 \mid X = 1) + P(\hat{y} = 1 \mid X = 1) P(Y = 0 \mid X = 1)\right), \\
&= 0.5 \cdot 0 + 0.5 \left(0.8 \cdot 0.2 + 0.2 \cdot 0.8\right), \\
&= 0.16.
\end{aligned}$$

Question 3

Given $X = 0$, suppose i choose $Y = 0$ with probability a, and $Y = 1$ with probability $1 - a$. Then the probability that i choose $Y = \hat{y}$ correctly, is given by:

$$P(Y = \hat{y} \mid X = 0) = a \cdot P(Y = 0 \mid X = 0) + (1 - a) P(Y = 1 \mid X = 0).$$

The expected 0-1 loss is then given by:

$$\begin{aligned}\mathcal{R}_p\left(\hat{y}\left(a\right)\right) &= 1 - P\left(Y = \hat{y} \mid X = 0\right), \\ &= 1 - \left(a \cdot P\left(Y = 0 \mid X = 0\right) + \left(1 - a\right) P\left(Y = 1 \mid X = 0\right)\right).\end{aligned}$$

I want to minimize the expected 0-1 loss:

$$\min_{a \in [0,1]} \mathcal{R}_p\left(\hat{y}\left(a\right)\right).$$

Since $\mathcal{R}_p\left(\hat{y}\left(a\right)\right)$ is linear, a value of a that minimizes $\mathcal{R}_p\left(\hat{y}\left(a\right)\right)$ will always be a boundary case, either $a^* = 0$ or $a^* = 1$: $\mathcal{R}_p\left(\hat{y}\left(a^*\right)\right) \leq \mathcal{R}_p\left(\hat{y}\left(a\right)\right)$ for all $a \in [0, 1]$. Therefore, the deterministic classification from question 1, where $y = \arg\max_{y \in \{0,1\}} P\left(Y = y \mid X = x\right)$, is a weakly optimal classifier when using the 0-1 loss as a measure for performance.

2 Logistic Regression (50 points)

2.1 Cross-entropy error measure (16 points)

2.1.1 Rewriting negative logarithmic likelihood as cross-entropy

In the case where $Y \in \{-1, 1\}$,and $h\left(\boldsymbol{x}\right) = P\left(Y = 1 \mid X = \boldsymbol{x}\right)$,i can write the data generation process:

$$\begin{aligned}P\left(Y = y \mid X = \boldsymbol{x}\right) &= \begin{cases} h(\boldsymbol{x}) & \text{for } y = 1 \\ 1 - h\left(\boldsymbol{x}\right) & \text{for } y = -1 \end{cases}, \\ &= \mathbf{1}_{\{y=1\}} h(\boldsymbol{x}) + \mathbf{1}_{\{y=-1\}} \left(1 - h(\boldsymbol{x})\right).\end{aligned}$$

From above i can derive the negative likelihood function:

$$\mathcal{L}\left(\boldsymbol{w}\right) = \frac{1}{N} \prod_{n=1}^{N} \left[-P\left(Y_n = y_n \mid X_n = \boldsymbol{x_n}; \boldsymbol{w}\right)\right].$$

The corresponding negative log-likelihood function is the:

$$\log\left(\mathcal{L}\left(\boldsymbol{w}\right)\right) = \frac{1}{N} \sum_{n=1}^{N} \log\left(\frac{1}{P\left(Y_n = y_n \mid X_n = \boldsymbol{x}_n; \boldsymbol{w}\right)}\right).$$

Substituting in $P\left(Y = y \mid X = \boldsymbol{x}\right) = \mathbf{1}_{\{y=1\}} h(\boldsymbol{x}) + \mathbf{1}_{\{y=-1\}} \left(1 - h(\boldsymbol{x})\right)$, yields:

$$\log\left(\mathcal{L}\left(\boldsymbol{w}\right)\right) = \frac{1}{N} \sum_{n=1}^{N} \left[\mathbf{1}_{\{y=1\}} \log\left(\frac{1}{h(\boldsymbol{x_n}; \boldsymbol{w})}\right) + \mathbf{1}_{\{y=-1\}} \log\left(\frac{1}{1 - h(\boldsymbol{x_n}; \boldsymbol{w})}\right)\right],$$

which is what i wanted to show. From now on i define:

$$\mathcal{L}\left(\boldsymbol{w}\right) = \frac{1}{N} \sum_{n=1}^{N} \left[\mathbf{1}_{\{y=1\}} \log\left(\frac{1}{h(\boldsymbol{x_n}; \boldsymbol{w})}\right) + \mathbf{1}_{\{y=-1\}} \log\left(\frac{1}{1 - h(\boldsymbol{x_n}; \boldsymbol{w})}\right)\right]. \tag{2}$$

2.1.2 Cross-entropy and logistic regression

To show that:

$$\mathcal{L}(\boldsymbol{w}) = \frac{1}{N}\sum_{n=1}^{N}\left[\mathbf{1}_{\{y=1\}}\log\left(\frac{1}{h(\boldsymbol{x_n};\boldsymbol{w})}\right) + \mathbf{1}_{\{y=-1\}}\log\left(\frac{1}{1-h(\boldsymbol{x_n};\boldsymbol{w})}\right)\right], \tag{2}$$

$$= \frac{1}{N}\sum_{n=1}^{N}\log\left(1+e^{-y_n\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_n}\right), \tag{1}$$

i use, that the sigmoid function has the following property:

$$P(Y=y \mid X=\boldsymbol{x};\boldsymbol{w}) = \theta\left(y\boldsymbol{w}^T\boldsymbol{x}\right),$$
$$= \begin{cases}\theta\left(\boldsymbol{w}^T\boldsymbol{x}\right) & \text{for } y=1\\ 1-\theta\left(\boldsymbol{w}^T\boldsymbol{x}\right) & \text{for } y=-1\end{cases},$$

where

$$\theta(z) = \frac{1}{1+e^{-z}}.$$

This means that:

$$-\log\left(\theta(z)\right) = \log\left(1+e^{-z}\right),$$
$$\Rightarrow -\log\left(\theta\left(y\boldsymbol{w}^T\boldsymbol{x}\right)\right) = \log\left(1+e^{-y\boldsymbol{w}^T\boldsymbol{x}}\right).$$

Plugging the above expression into equation (1) from the assignment, and using that $P(Y=y \mid X=\boldsymbol{x};\boldsymbol{w}) = \theta\left(y\boldsymbol{w}^T\boldsymbol{x}\right)$, yields:

$$\mathcal{L}(\boldsymbol{w}) = \frac{1}{N}\sum_{n=1}^{N} -\log\left(\theta\left(y_n\boldsymbol{w}^T\boldsymbol{x}_n\right)\right),$$
$$= \frac{1}{N}\sum_{n=1}^{N} -\log\left(P(Y=y_n \mid X=\boldsymbol{x}_n;\boldsymbol{w})\right),$$
$$= \frac{1}{N}\sum_{n=1}^{N}\left[\mathbf{1}_{\{y=1\}}\log\left(\frac{1}{h(\boldsymbol{x_n};\boldsymbol{w})}\right) + \mathbf{1}_{\{y=-1\}}\log\left(\frac{1}{1-h(\boldsymbol{x_n};\boldsymbol{w})}\right)\right],$$

which is what i wanted to show.

2.2 Logistic regression loss gradient (24 points)

2.2.1 Gradient for -1 and 1

Consider the negative log-likelihood function (eq. (1)):

$$\mathcal{L}(\boldsymbol{w}) = \frac{1}{N}\sum_{n=1}^{N}\log\left(1+e^{-y_n\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}_n}\right).$$

Using the chain rule, the gradient of the negative log-likelihood can be derived:

$$
\begin{aligned}
\nabla \mathcal{L}(\boldsymbol{w}) &= \frac{\partial \mathcal{L}(\mathbf{w})}{\partial \boldsymbol{w}} \\
&= \frac{1}{N} \sum_{n=1}^{N} \begin{pmatrix} \frac{\partial}{\partial w_1} \log\left(1 + e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_n}\right) \\ \frac{\partial}{\partial w_2} \log\left(1 + e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_n}\right) \\ \vdots \\ \frac{\partial}{\partial w_d} \log\left(1 + e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_n}\right) \end{pmatrix}, \\
&= \frac{1}{N} \sum_{n=1}^{N} \begin{pmatrix} -y_n x_{n1} \frac{e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_{n1}}}{1+e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_{n1}}} \\ -y_n x_{n2} \frac{e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_{n2}}}{1+e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_{n2}}} \\ \vdots \\ -y_n x_{n2} \frac{e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_{nd}}}{1+e^{-y_n \boldsymbol{w}^{\mathrm{T}} x_{nd}}} \end{pmatrix}, \\
&= \frac{1}{N} \sum_{n=1}^{N} -y_n \boldsymbol{x}_n \frac{e^{-y_n \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n}}{1 + e^{-y_n \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n}}, \\
&= \frac{1}{N} \sum_{n=1}^{N} -y_n \boldsymbol{x}_n \frac{e^{-y_n \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n} e^{y_n \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n}}{\left(1 + e^{-y_n \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n}\right) e^{1 y \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n}}, \\
&= \frac{1}{N} \sum_{n=1}^{N} -y_n \boldsymbol{x}_n \frac{1}{1 + e^{y_n \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n}}, \\
&= \frac{1}{N} \sum_{n=1}^{N} -y_n \boldsymbol{x}_n \theta\left(-y_n \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_n\right).
\end{aligned}
$$

2.2.2 Gradient for 0 and 1

When $Y \in \{0, 1\}, P(Y = y \mid X = \boldsymbol{x}; \boldsymbol{w})$ can be written:

$$
\begin{aligned}
P(Y = y \mid X = \boldsymbol{x}; \boldsymbol{w}) &= \theta\left(\boldsymbol{w}^T \boldsymbol{x}\right)^y \left(1 - \theta\left(\boldsymbol{w}^T \boldsymbol{x}\right)\right)^{1-y}, \\
&= \begin{cases} \theta\left(\boldsymbol{w}^T \boldsymbol{x}\right) & \text{for } y = 1 \\ 1 - \theta\left(\boldsymbol{w}^T \boldsymbol{x}\right) & \text{for } y = 0 \end{cases}.
\end{aligned}
$$

Using the above expression, the negative log-likelihood function can be written:

$$
\begin{aligned}
\mathcal{L}(\boldsymbol{w}) &= -\frac{1}{N}\sum_{n=1}^{N}\log\left(P\left(Y=y_n \mid X=\boldsymbol{x_n};\boldsymbol{w}\right)\right), \\
&= -\frac{1}{N}\sum_{n=1}^{N}\log\left(\theta\left(\boldsymbol{w}^T\boldsymbol{x}_n\right)^{y_n}\left(1-\theta\left(\boldsymbol{w}^T\boldsymbol{x}_n\right)\right)^{1-y_n}\right), \\
&= -\frac{1}{N}\sum_{n=1}^{N} y_n\log\left(\theta\left(\boldsymbol{w}^T\boldsymbol{x}_n\right)\right)+(1-y_n)\log\left(1-\theta\left(\boldsymbol{w}^T\boldsymbol{x}_n\right)\right), \\
&= -\frac{1}{N}\sum_{n=1}^{N}\left(y_n\log\left(\frac{1}{1+e^{-\boldsymbol{w}^T\boldsymbol{x}_n}}\right)+(1-y_n)\log\left(1-\frac{1}{1+e^{-\boldsymbol{w}^T\boldsymbol{x}_n}}\right)\right), \\
&= -\frac{1}{N}\sum_{n=1}^{N}\left(-y_n\log\left(1+e^{-\boldsymbol{w}^T\boldsymbol{x}_n}\right)+(1-y_n)\log\left(\frac{e^{-\boldsymbol{w}^T\boldsymbol{x}_n}}{1+e^{-\boldsymbol{w}^T\boldsymbol{x}_n}}\right)\right), \\
&= -\frac{1}{N}\sum_{n=1}^{N}\left(-y_n\log\left(1+e^{-\boldsymbol{w}^T\boldsymbol{x}_n}\right)+(1-y_n)\log\left(\frac{1}{1+e^{\boldsymbol{w}^T\boldsymbol{x}_n}}\right)\right) \\
&= -\frac{1}{N}\sum_{n=1}^{N}\left(-y_n\log\left(1+e^{-\boldsymbol{w}^T\boldsymbol{x}_n}\right)-(1-y_n)\log\left(1+e^{\boldsymbol{w}^T\boldsymbol{x}_n}\right)\right), \\
&= \frac{1}{N}\sum_{n=1}^{N}\left(y_n\log\left(1+e^{-\boldsymbol{w}^T\boldsymbol{x}_n}\right)+(1-y_n)\log\left(1+e^{\boldsymbol{w}^T\boldsymbol{x}_n}\right)\right).
\end{aligned}
$$

The gradient can then be derived:

$$
\begin{aligned}
\nabla\mathcal{L}(\boldsymbol{w}) &= \frac{\partial\mathcal{L}(\mathbf{w})}{\partial\boldsymbol{w}} \\
&= \frac{1}{N}\sum_{n=1}^{N}\begin{pmatrix}
\frac{\partial}{\partial w_1}\left(y_n\log\left(1+e^{-\boldsymbol{w}^T x_n}\right)+(1-y_n)\log\left(1+e^{\boldsymbol{w}^T x_n}\right)\right) \\
\frac{\partial}{\partial w_2}\left(y_n\log\left(1+e^{-\boldsymbol{w}^T x_n}\right)+(1-y_n)\log\left(1+e^{\boldsymbol{w}^T x_n}\right)\right) \\
\vdots \\
\frac{\partial}{\partial w_d}\left(y_n\log\left(1+e^{-\boldsymbol{w}^T x_n}\right)+(1-y_n)\log\left(1+e^{\boldsymbol{w}^T x_n}\right)\right)
\end{pmatrix}, \\
&= \frac{1}{N}\sum_{n=1}^{N}\begin{pmatrix}\nabla l_1\\ \nabla l_2\\ \vdots\\ \nabla l_d\end{pmatrix}
\end{aligned}
$$

For the sake of simplicity, i will derive ∇l_i, and then generalize to $\nabla \mathcal{L}(\boldsymbol{w}) = \frac{1}{N}\sum_{n=1}^{N} \nabla \boldsymbol{l}$:

$$\begin{aligned}
\nabla l_i &= \frac{\partial}{\partial w_i}\left(y_n \log\left(1+e^{-\boldsymbol{w}^T x_{ni}}\right) + (1-y_n)\log\left(1+e^{\boldsymbol{w}^T x_{ni}}\right)\right), \\
&= -x_{ni} y_n \frac{e^{-\boldsymbol{w}^T x_{ni}}}{1+e^{-\boldsymbol{w}^T x_{ni}}} + x_{ni}(1-y_n)\frac{e^{\boldsymbol{w}^T x_{ni}}}{1+e^{\boldsymbol{w}^T x_{ni}}}, \\
&= -x_{ni} y_n \left(1 - \frac{1}{1+e^{-\boldsymbol{w}^T x_{ni}}}\right) + x_{ni}(1-y_n)\frac{1}{1+e^{-\boldsymbol{w}^T x_{ni}}}, \\
&= -x_{ni}\left(y_n - y_n \frac{1}{1+e^{-\boldsymbol{w}^T x_{ni}}} - \frac{1}{1+e^{-\boldsymbol{w}^T x_{ni}}} + y_n \frac{1}{1+e^{-\boldsymbol{w}^T x_{ni}}}\right), \\
&= -x_{ni}\left(y_n - \frac{1}{1+e^{-\boldsymbol{w}^T x_{ni}}}\right).
\end{aligned}$$

Plugging the above expression into $\nabla \mathcal{L}(\boldsymbol{w})$, yields:

$$\begin{aligned}
\nabla \mathcal{L}(\boldsymbol{w}) &= \frac{1}{N}\sum_{n=1}^{N} \begin{pmatrix} -x_{n1}\left(y_n - \frac{1}{1+e^{-\boldsymbol{w}^T x_{n1}}}\right) \\ -x_{n2}\left(y_n - \frac{1}{1+e^{-\boldsymbol{w}^T x_{n2}}}\right) \\ \vdots \\ -x_{nd}\left(y_n - \frac{1}{1+e^{-\boldsymbol{w}^T x_{nd}}}\right) \end{pmatrix}, \\
&= -\frac{1}{N}\sum_{n=1}^{N} \begin{pmatrix} \left(y_n - \theta\left(\boldsymbol{w}^T x_{n1}\right)\right) x_{n1} \\ \left(y_n - \theta\left(\boldsymbol{w}^T x_{n2}\right)\right) x_{n2} \\ \vdots \\ \left(y_n - \theta\left(\boldsymbol{w}^T x_{nd}\right)\right) x_{nd} \end{pmatrix}, \\
&= -\frac{1}{N}\sum_{n=1}^{N}\left(y_n - \theta\left(\boldsymbol{w}^T \boldsymbol{x}_n\right)\right)\boldsymbol{x}_n.
\end{aligned}$$

2.2.3 Influence of misclassified examples

For $\boldsymbol{Y} \in \{0, 1\}$, the gradient contribution of one observation is given by:

$$\nabla l_{n=i} = -\frac{1}{N}\left(y_i - \theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right)\right)\boldsymbol{x}_i.$$

I start by defining $m_i = y_i - \theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right)$.

The sample can be misclassified in two ways: If the model predicts $\hat{y}_i = 0$, when $y_i = 1$, then $\theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right) < 0.5$,which yields:

$$m_i = 1 - \theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right) > 0.5.$$

If the model predicts $\hat{y}_i = 1$, when $y_i = 0$, then $\theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right) > 0.5$, which yields:

$$m_i = 0 - \theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right) < -0.5.$$

Thus, the gradient contribution when a sample is misclassified is:

$$|m_i^{mis}| = |y_i - \theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right)| > 0.5.$$

When the sample is correctly classified (ie. $\hat{y}_i = y_i$) then:

$$|m_i^{cor}| = |y_i - \theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right)| < 0.5,$$

since $\theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right) > 0.5$,when $y_i = 1$, and $\theta\left(\boldsymbol{w}^T \boldsymbol{x}_i\right) < 0.5$,when $y_i = 0$.

This means, that the gradient contribution of a misclassified sample is always bigger than a correctly classified sample (for a given sample):

$$|\nabla l_{n=i}^{mis}| = |-\frac{1}{N}\left(m_i^{mis}\right)\boldsymbol{x}_i| > |\nabla l_{n=i}^{cor}| = |-\frac{1}{N}\left(m_i^{cor}\right)\boldsymbol{x}_i|.$$

For $\boldsymbol{Y} \in \{-1, 1\}$, the gradient contribution of one observation is given by:

$$\nabla l_{n=i} = -\frac{1}{N} y_i \boldsymbol{x}_i \theta\left(-y_i \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_i\right).$$

I first define $m_i = -y_i \boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_i$.
If the model predicts $\hat{y}_i = 1$, when $y_i = -1$, then:

$$\boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_i > 0 \Rightarrow m_i > 0.$$

If the model predicts $\hat{y}_i = -1$, when $y_i = 1$, then:

$$\boldsymbol{w}^{\mathrm{T}} \boldsymbol{x}_i < 0 \Rightarrow m_i > 0.$$

Thus:

$$m_i^{mis} > 0.$$

If the model classifies $\hat{y}_i = y_i$ correctly, then m_i becomes negative:

$$m_i^{cor} < 0.$$

Since the sigmpid $\theta\left(m_i\right)$ is increasing in m_i, the gradient contribution of a given misclasiffied sample will always be bigger than the correctly classified sample:

$$\begin{aligned} m_i^{cor} < m_i^{mis} \Rightarrow \theta\left(m_i^{cor}\right) < \theta\left(m_i^{mis}\right), \\ \Rightarrow |\nabla l_{n=i}^{cor}| = |-\frac{1}{N} y_i \boldsymbol{x}_i \theta\left(m_i^{cor}\right)| < |\nabla l_{n=i}^{mis}| = |-\frac{1}{N} y_i \boldsymbol{x}_i \theta\left(m_i^{cor}\right) \end{aligned}$$

2.3 Log-odds (10 points)

To show that:

$$f(\mathbf{x}) = \sigma(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b) = P(Y = 1 \mid X = \boldsymbol{x}),$$

where $\sigma(\cdot)$is the logistic function $\sigma\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b\right) = \frac{1}{1+e^{-\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b\right)}}$, i start by rewriting eq. (4) from the assignment:

$$\begin{aligned}
\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b &= \log\left(\frac{P(Y = 1 \mid X = x)}{P(Y = 0 \mid X = x)}\right), \\
&= -\log\left(\frac{P\left(Y = 0 \mid X = x\right)}{P\left(Y = 1 \mid X = x\right)}\right), \\
\Leftrightarrow -\left(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b\right) &= \log\left(\frac{P\left(Y = 0 \mid X = x\right)}{P\left(Y = 1 \mid X = x\right)}\right), \\
\Leftrightarrow e^{-(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b)} &= \frac{P\left(Y = 0 \mid X = x\right)}{P\left(Y = 1 \mid X = x\right)}, \\
&= \frac{1 - P\left(Y = 1 \mid X = x\right)}{P\left(Y = 1 \mid X = x\right)}, \\
&= \frac{1}{P\left(Y = 1 \mid X = x\right)} - 1, \\
\Leftrightarrow 1 + e^{-(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b)} &= \frac{1}{P\left(Y = 1 \mid X = x\right)}, \\
\Leftrightarrow P\left(Y = 1 \mid X = x\right) &= \frac{1}{1 + e^{-(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b)}}.
\end{aligned}$$

Comparing the expression above to eq. (3) from the assignment, it is obvious that $\sigma(\cdot)$ is the logistic function $\sigma(z) = \frac{1}{1+e^{-z}}$:

$$f(\mathbf{x}) = \sigma(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} + b) = P(Y = 1 \mid X = \boldsymbol{x}) = \frac{1}{1 + e^{-(\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x}+b)}}.$$

3 Sleep Well (34 points)

Note to self: description of software used should include parameters values etc.

3.1 Data understanding and preprocessing (4 points)

I calculated the frequency of classes in the training data using python (numpy and pandas). The results are shown in the table below:

Class	Count	Frequency
0	17.566	0.521
1	3.221	0.096
2	8.523	0.253
3	1.583	0.047
4	2.831	0.084
Total:	**33.724**	**1.000**

3.2 Classification (30 points)

3.2.1 Logistic regression

I used python to apply the multinomial logistic regression (numpy and scikit-learn). I used stochastic gradient descent (SGD) to fit the model, with a tolerance of $1 \cdot 10^{-8}$. I did not use any regularization as using two-norm regularization didn't improve the 0-1 loss meaningfully for any values of C.

Using logistic regression with no regularization, i got a training 0-1 loss of $L_{train}(h) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}[h(\boldsymbol{x}_{train,i}) \neq y_{train,i}] \approx 0.1495$,and a test 0-1 loss of $L_{test}(h) \approx 0.0956$.

3.2.2 Random forest

I used python to apply the random forests clasiffier algorithm (numpy and scikit-learn). I used Gini index as the impurity measure and chose the maximum variables considered per split: $m = \sqrt{D} = \sqrt{16} = 4$. I chose the minimum required samples a node must have to split further as 5.

The OOB error, training and test loss (0–1 loss) for each of the three random forests are shown in the table below:

Number of trees (B)	OOB error	Train loss (0–1)	Test loss (0–1)
50	0.8465	0.0080	0.1131
100	0.8508	0.0063	0.1122
200	0.8534	0.0050	0.1098

The random forest with 200 trees performed slightly better compared to 50 and 100 trees (less than 1 pct. point decrease on 0–1 test loss). Compared to the increase in required compute, this small performance gain may not be worth the extra compute depending on what the model is needed for.

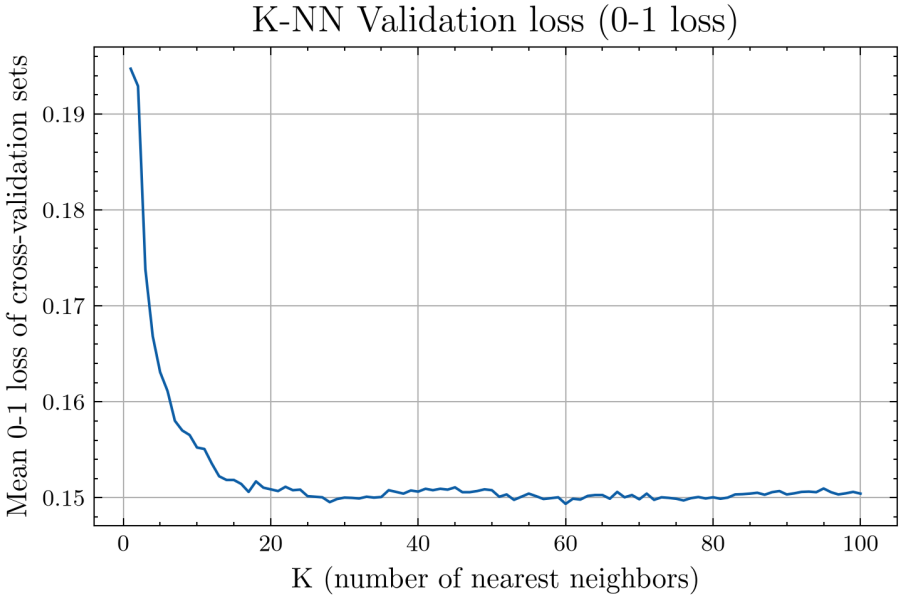
3.2.3 Nearest neighbor

I used python to apply K-Nearest neighbor (numpy and scikit-learn). I used the euclidean norm as the distance metric with no regularization.

I used cross-validation with five validation sets to determine the number of neighbors (K). I used the mean 0-1 loss of the validation sets for each $K \in \{1, 2, \cdots, 100\}$ as measure of the predictive performance of K. Using the method i found, that $K = 60$ gave the minimum mean 0-1 loss with $L_{validation} \approx 0.1494$.

Using $K = 60$ i calculated the training 0-1 loss: $L_{train} \approx 0.1462$, and the test 0-1 loss: $L_{test} \approx 0.0980$. Surprisingly the test 0-1 loss was the lowest of all the 0-1 losses by quiet a lot.

The figure below shows the mean 0-1 loss of the validations sets as a function of K:



Looking at the figure, it seems the 0-1 loss is almost constant for $K \geq 25$, with a 0-1 loss around 0.15. I would likely have gotten close to the same performance using $K = 25$ rather than $K = 60$.